Reg. No. | | | | | | | | | | |

[22ESEC13]

FIRST SEMESTER B.Tech EXAMINATION, FEBRUARY - 2023

ANALOG AND DIGITAL CIRCUITS-1

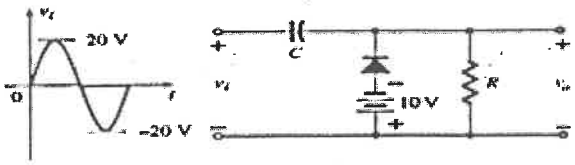
Time: 3 Hours **Maximum Marks: 100**

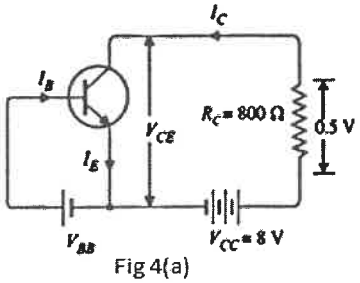
Instructions: **i.** Missing data may be suitably assumed.

ii. Answer All Questions

PART - A

5 X 2=10

1.	**a.**	Why BJT is called current controlled device?	L2	CO1	**02**
	b.	Why FET is called voltage controlled device?	L2	CO1	**02**
	c.	When does a transistor works as an amplifier?	L3	CO2	**02**
	d.	Which gates are called as the universal gates? What are its advantages?	L3	CO3	**02**
	e.	State the distributive property of Boolean algebra.	L2	CO3	**02**
		PART - B			**6X 5=30**
2.	**a.**	Explain the effect of temperature on the forward and reverse characteristics of a PN junction diode, with neat sketches.	L2	CO1	**05**
		OR			
	b.	Show that half wave rectifier efficiency is 40.1%	L3	CO2	**05**
3.	**a.**	Determine the upper & lower levels of the output voltage for the circuit shown in fig. when the input voltage is ±20 V Also calculate the capacitor voltage.  Fig 3(a)	L3	CO2	**05**
		OR			
	b.	Draw the circuit diagram of a bridge rectifier and explain its working. Plot its input and output waveforms.	L3	CO2	**05**
4.	**a.**	Define the current gain α and β. Derive the relation between them.	L3	CO2	**05**
		OR			
	b.	A transistor is connected in common emitter (CE) configuration in which collector supply is 8 V and the voltage drop across resistance R_C connected in the collector circuit is 0.5 V. The value of R_C = 800 Ω. If α = 0.96, determine:	L3	CO2	**05**

		i. Collector-emitter voltage ii. Base current.  Fig 4(a)			
5.	**a.**	Sketch the output static characteristic curves of a npn transistor in CB configuration. Explain the output characteristics for different regions of operation.	L2	CO2	**05**
		OR			
	b.	Explain in detail the drain characteristics of NJFET	L2	CO2	**05**
6.	**a.**	Explain in detail the transfer characteristics of NJFET	L2	CO1	**05**
		OR			
	b.	What are the advantages of the FET over the conventional transistors? What do the terms Unipolar and Bipolar refer to?	L2	CO1	**05**
7.	**a.**	Explain the working of SCR using crystal structure and their three operating modes.	L2	CO1	**05**
		OR			
	b.	Explain the static characteristics of TRIAC	L2	CO1	**05**
		PART-C **6 X 10=60**			
8.	**a.**	**i.** Explain UJT as a relaxation Oscillator. **ii.** Convert octal number $(420.23)_8$ to decimal number.	L2	CO1	**6+4**
		OR			
	b.	**i.** Find the decimal equivalent of $(111001101)_2$ **ii.** Explain Gray code conversion and list its significance.	L3	CO3	**5+5**
9.	**a.**	**i.** Realize three basic gates using NOR gates **ii.** State and explain De-Morgan's Laws	L3	CO3	**6+4**
		OR			
	b.	**i.** Explain sum of Product (SOP) forms with an example. **ii.** Convert the Boolean expression into standard SOP form. X=A+B'C	L2	CO3	**4+6**

10.	**a.**	**i.** Prove that A+A'B= A+B **ii.** Simplify the following SOP expression using K-Map X= A'BC'D'+ABC'D'+A'BC'D+ABC'D+A'BCD+ABCD	L4	CO3	**3+7**
		OR			
	b.	Explain half and full subtractor with their truth table, logic diagram and their Boolean expression.	L3	Co3	**4+6**
11.	**a.**	What is Decoder? Explain 3 to 8 line Decoder with logic diagram.	L2	CO4	**10**
		OR			
	b.	What is Demultiplexers (DEMUX)? Explain 1:4 DEMUX with their function table and Logic diagram	L2	CO4	**10**
12.	**a.**	**i.** What is magnitude comparator and explain one bit magnitude comparator with logic diagram and function table **ii.** Convert the following binary numbers to hexadecimal number **(i)** 00110111 **(ii)** 111101010110	L3	CO4	**6+4**
		OR			
	b.	**i.** What are the differences between combinational and sequential circuits? Explain SR Latch using NOR Gate. **ii.** Differentiate between synchronous and asynchronous counters.	L3	CO4	**7+3**
13.	**a.**	Explain 3 bit asynchronous(ripple) counter with wave form.	L3	CO5	**10**
		OR			
	b.	**i.** Explain the drain characteristics of enhancement mode NMOSFET. **ii.** Compare CB,CE and CC configurations of BJT.	L3	CO2	**5+5**